

MEALOPTIMIZA — WHATSAPP WEBHOOK ARCHITECTURE

Step-by-Step Implementation Guide for AI Food Vision & Diaspora Parent Care

1. Executive Overview & Technical Blueprint

The MealOptimiza WhatsApp integration enables zero-friction dietary and metabolic tracking. Users (especially elderly parents in Nigeria or Ghana under the **Diaspora Parent Care** subscription) can simply snap a photo of their meal, send a voice note, or text '*just ate pounded yam and egusi with one titus fish*' directly on WhatsApp. The system processes the entry in real time, logs it to the clinical diary, and returns an instant glycemic analysis.

End-to-End Event Lifecycle:

- 1. Ingestion:** Parent snaps photo of food on WhatsApp \$to\$ Sent to MealOptimiza dedicated business number.
- 2. Webhook Event:** Meta Cloud API or Twilio triggers an HTTPS POST webhook to your server endpoint.
- 3. Media Download:** Server fetches image using temporary media access token \$to\$ Passes to AI Vision Pipeline.
- 4. Clinical Analysis:** Vision model extracts African staples, portions, calories, carbs, fiber, and glycemic surge.
- 5. Database Sync:** Saved to Supabase/PostgreSQL under user's profile with timestamp & nutrients.
- 6. Instant WhatsApp Reply:** Parent receives Avo Mascot scorecard: '*Plate Score: 85%! Low spike risk*'.
- 7. Diaspora Alert:** If dangerous glucose surge detected, an instant notification is sent to the child in London/US.

2. Choosing Your WhatsApp Webhook Provider

There are two primary industry routes to power the MealOptimiza WhatsApp bot. Here is the decision matrix:

Criteria	Option A: Meta Cloud API (Official)	Option B: Twilio for WhatsApp
Best For	Production scaling & lowest ongoing cost	Immediate 10-minute testing & sandbox demo
Cost per Chat	First 1,000 service chats/mo are 100% FREE. Then ~\$0.005–\$0.01 per conversation.	~\$0.005 Twilio fee + Meta WhatsApp conversation rate (~\$0.01–\$0.02 total).
Setup Friction	Requires Meta Business Manager & phone number.	Instant pre-configured Sandbox (no verification needed).
Recommendation	Primary Engine for public production launch.	Testing Engine while awaiting Meta verification.

3. Step-by-Step: Meta Cloud API Setup (Official Route)

Follow these 5 sequential steps to configure your official WhatsApp Business Webhook through Meta:

Step 3.1: Create Meta Developer Account & App

- Navigate to **developers.facebook.com** and log in with your primary Facebook account.
- Click **My Apps** \$to\$ **Create App**.
- Select **Other** \$to\$ Select **Business** as the app type.
- Enter App Name: **MealOptimiza AI Assistant** \$to\$ Link to your Meta Business Portfolio.

Step 3.2: Add WhatsApp Product to Your App

- In the App Dashboard left sidebar, find **Add Products** \$to\$ Click **Set Up** on **WhatsApp**.
- Meta will automatically generate a **Test Phone Number** and a **Temporary 24-Hour Access Token**.
- In **To**, enter your personal WhatsApp number and send a test template message to confirm connectivity.

Step 3.3: Set Up a Permanent System User Access Token

Crucial: Temporary tokens expire in 24 hours. You need a permanent system token for production:

1. Go to **business.facebook.com/settings** \$to\$ **System Users** \$to\$ Click **Add**.
2. Name: mealoptimiza-whatsapp-bot \$to\$ Role: **Admin**.
3. Click **Generate New Token** \$to\$ Select your MealOptimiza App.
4. Select Permissions: whatsapp_business_messaging and whatsapp_business_management.
5. Copy this permanent token into your server .env as WHATSAPP_CLOUD_API_TOKEN.

Step 3.4: Configure Webhook Verification Handshake

Meta requires a **GET** verification handshake before it will dispatch messages to your server. Your server must respond with the hub.challenge parameter when verified.

```
// GET /api/webhooks/whatsapp - Meta Handshake Verification
export async function GET(req: Request) {
  const { searchParams } = new URL(req.url);
  const mode = searchParams.get("hub.mode");
  const token = searchParams.get("hub.verify_token");
  const challenge = searchParams.get("hub.challenge");

  if (mode === "subscribe" && token === process.env.WHATSAPP_VERIFY_TOKEN) {
    console.log("■ WhatsApp Webhook Verified by Meta!");
    return new Response(challenge, { status: 200 });
  }
  return new Response("Forbidden", { status: 403 });
}
```

Step 3.5: Subscribe to Webhook Fields in Meta Console

- In Meta Developer Console \$to\$ WhatsApp \$to\$ **Configuration** \$to\$ **Webhook**.
- Enter Callback URL: <https://api.yourdomain.com/api/webhooks/whatsapp>
- Enter Verify Token: (Matches your secret WHATSAPP_VERIFY_TOKEN in .env)
- Click **Verify and Save**.
- In Webhook Fields table, click **Subscribe** next to **messages**.

4. Ingestion Pipeline: Handling Food Photos, Text & Voice

When a user texts or sends an image, Meta dispatches a **POST** event containing the payload. Here is the production implementation structure:

```
// POST /api/webhooks/whatsapp - Incoming Message Handler
export async function POST(req: Request) {
  const body = await req.json();
  const entry = body.entry?.[0]?.changes?.[0]?.value;
  const message = entry?.messages?.[0];
  if (!message) return new Response("EVENT_RECEIVED", { status: 200 });

  const senderNumber = message.from; // e.g. "2348012345678" or "447123456789"

  if (message.type === "image") {
    const imageId = message.image.id;
    // 1. Fetch image URL from Meta Graph API using imageId
    // 2. Stream image buffer to Gemini/GPT-4o Vision API
    // 3. Return African Glycemic Breakdown & Plate Score
  } else if (message.type === "text") {
    const textBody = message.text.body;
    // Parse meal description (e.g. "2 scoops amala with gbegiri and ewedu")
  }

  return new Response("OK", { status: 200 });
}
```

5. AI Food Vision & Clinical Analysis Pipeline

To provide instant clinical value to the user on WhatsApp, pipe the received image or text into an AI model with this specialized prompt:

Clinical African Nutrition Vision Prompt:

'You are Avo, the certified African metabolic dietitian for MealOptimiza. Analyze this meal photo/text. Identify African staples (Amala, Fufu, Eba, Jollof, Yam, Waakye, Sadza), soups (Efo Riro, Egusi, Okra, Bitter Leaf), proteins, and estimated portions. Calculate: Estimated Calories, Carbs(g), Protein(g), Fiber(g), and Glycemic Surge Risk (Low, Medium, High). Format the reply for WhatsApp with clean emojis, 9-inch plate advice, and 1 clinical action habit.'

Sample WhatsApp Output Returned to Parent:

[illegible]

6. Diaspora Parent Care Alert Engine

When an elderly parent logs a meal with dangerous glycemic risk (e.g. 3 huge scoops of white garri with sugary malt beverage), the webhook triggers an automated alert to the sponsoring family member:

Automated Family Notification Rules:

- **Trigger Condition:** Carbs > 85g AND Glycemic Impact = 'HIGH', or logged glucose > 180 mg/dL.
- **Action:** Dispatches email & push notification to Diaspora subscriber: *'Hi Tunde, your mum just logged a High-Spike meal in Lagos. Avo recommended adding a 15-minute walk and 2 cups of water.'*
- **Weekly Digest:** Every Sunday at 6 PM, the system compiles a 7-day Clinical PDF Report and emails it to the Diaspora subscriber.

7. Fast-Track Testing: Twilio WhatsApp Sandbox (10 Mins)

If you want to demo or test this today while Meta business verification is pending:

1. Create an account at [twilio.com](https://www.twilio.com) Go to **Messaging** Try **WhatsApp**.
2. Twilio provides a sandbox number (e.g. +1 415 523 8886).
3. Send join [sandbox-keyword] from your phone to connect.
4. In Twilio Sandbox Settings Set **'When a message comes in'** to your Webhook URL.
5. Twilio automatically delivers text, media URLs, and sender info via standard POST body.

8. Production Security & Policy Guidelines

- **24-Hour Customer Service Window:** When a user texts your bot, you have 24 hours to reply with free-form AI messages. Outside 24 hours, you must use pre-approved Meta Template messages.
- **Signature Validation:** Always verify the X-Hub-Signature-256 header using your App Secret to ensure incoming webhook calls genuinely originate from Meta.

- **Idempotency:** Store processed message.id values in Redis or PostgreSQL for 24 hours to ignore duplicate webhook retry requests from Meta during network latency.
- **Phone Number Requirement:** For official Meta production, purchase a clean SIM or VoIP number not currently registered to any personal WhatsApp account.